

TRISTAN THAKUR

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, College of Computing
B.S. in Computer Science

Atlanta, Georgia
May 2026

- Faculty Honors, Dean's List, Concentration in AI, Modeling, & Simulation

GPA: 4.0/4.0

SKILLS

Programming: C++, Python, Java, C, R, SQL, x86 Assembly, Bash

Tools: Linux, Git, Docker, Kubernetes, ROS, AWS, NumPy, PyTorch, Conda, Jupyter, MySQL, Conan, Poetry, Slurm, GDB

Languages: English (native), French (intermediate), Spanish (intermediate)

EXPERIENCE

SHIELD AI

Washington, DC

Autonomy Engineer Intern

June 2025 - August 2025

- Designed and deployed a path-planning framework in C++ for Group 5 fixed-wing UAS, producing kinematically flyable trajectories **3.47×** faster with a **+22 pp** success rate at a 2.0 s horizon.
- Developed a comprehensive stress-testing suite to rigorously evaluate the path-planning algorithm in randomized, high-complexity environments, uncovering multiple failure modes in multi-dimensional state space navigation.
- Built parallelized batch-testing infrastructure to filter noise from performance metrics and enable statistical benchmarking of planner variants; identified point of diminishing returns, reducing real-time planning latency by **0.5s**.
- Diagnosed a critical production bug that blocked agent exploration in complex obstacle fields, and rapidly implemented a fix adopted by **70+ engineers** across the autonomy stack.

GEORGIA TECH RESEARCH INSTITUTE

Atlanta, Georgia

Machine Learning Engineer Intern

May 2024 - December 2024

- Developed an AutoML pipeline in Python using an evolutionary Neural Architecture Search (NAS) algorithm to automate the construction of state-of-the-art airborne object detection models, parallelizing computationally intensive model evaluations across GPU nodes on an HPC cluster.
- Trained a Kolmogorov-Arnold Network as a surrogate evaluator to accurately predict model performance, eliminating **1100+ hours** of training computation for candidate architectures on 11 terabytes of high-speed aerial images.
- Compressed VGG-16 into a lightweight student classifier using knowledge distillation for the DoD AARO's fielded 'GREMLIN' autonomous ISR system, retaining comparable accuracy with just **0.1%** of the original parameters.

GEORGIA TECH STUDENT FOUNDATION INVESTMENTS COMMITTEE

Atlanta, Georgia

Head of Quantitative Research

January 2023 - December 2025

- Directed quantitative research for a **\$2M+** AUM university endowment fund, developing systematic trading models, signal frameworks, and portfolio optimization tools.
- Developed a Gaussian Hidden Markov Model for latent market regime detection, modeling volatility clustering and transition dynamics using unsupervised EM-based training and state probability inference.
- Implemented a Kalman Filter for real-time intraday beta estimation and dynamic hedge ratio calibration across equity exposures.

GEORGIA TECH VERTICALLY INTEGRATED PROJECTS

Atlanta, Georgia

Undergraduate Researcher

April 2023 - December 2024

- Led **9 students** on a Google & BMW-sponsored AI research team developing an evolutionary algorithm for generating high-performance image classification neural networks, surpassing CIFAR-10 benchmarks by **3.5%**.
- Engineered network branching mechanisms for residual/skip connections that increased classification F1 scores by **12%**.
- Improved generated models' collective performance by **9.5%** by integrated custom vision transformer layer with patch embedding & multi-headed self-attention into the architectural primitive set.

PROJECTS

Cellular Automata-LLM Ecological Policy Modeling & Simulation Framework

August 2025 - December 2025

- Architected a spatially explicit, stage-structured Cellular Automata simulator modeling invasive Kudzu dynamics with stochastic mortality, density-dependent diffusion, and probabilistic long-range spread across a land-cover grid.
- Built the LLM-simulation integration layer, defining structured state aggregation and budget-constrained action interfaces to enable adaptive policy control within a constrained, state-dependent dynamical environment.

Learning UAV State Estimation in GPS-Denied Environments

January 2025 - April 2025

- Architected a multimodal, learning-based state estimator for full 12-DoF UAV dynamics by fusing PCA-compressed ResNet18 visual embeddings with synchronized IMU and motor thrust signals over temporal flight windows.
- Benchmarked linear, feedforward, and recurrent architectures for nonlinear flight inference, achieving **~60%** reduction in state MAE and **sub-10cm** positional error in aggressive, GPS-denied flight regimes.